CSA1713 – Artificial Intelligence

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Slot – C

10. A* ALGORITHM

Code:

```
print("B. Deepthi Reddy")
print(192312288)
print("A* ALGORITHM")
graph = {}
n = int(input("Enter number of nodes: "))
for i in range(n):
    node = input("Node: ")
    graph[node] = {}
    e = int(input("Edges from " + node + ": "))
    for j in range(e):
        v = input("Connected node: ")
        c = int(input("Cost: "))
        graph[node][v] = c
h = {}
for i in graph:
    h[i] = int(input("Heuristic of " + i + ": "))
```

```
start = input("Start node: ")
goal = input("Goal node: ")

open = [start]
cost = {start: 0}
parent = {start: None}

while open:
    x = min(open, key=lambda i: cost[i] + h[i])
    open.remove(x)

    if x == goal:
        path = []

        while x:
            path.append(x)
            x = parent[x]

        print("Path:", path[::-1])
        break

    for y in graph[x]:
        g = cost[x] + graph[x][y]

        if y not in cost or g < cost[y]:
```

```
cost[y] = g
parent[y] = x

if y not in open:
    open.append(y)
```

Output: